$$2B + 3X_2 \xrightarrow{\Delta} 2BX_3$$

Formation of boron nitride:

Boron burns with dinitrogen at high temperatures to form boron nitride.

$$2B + N_2 \xrightarrow{\Delta} 2BN$$

Formation of oxides:

When boron is heated with oxygen around 900 K, it forms its oxide.

$$4B + 3O_2 \xrightarrow{900\ K} 2B_2O_3$$

Reaction with acids and alkali:

Halo acids have no reaction with boron. However, boron reacts with oxidising acids such as sulphuric acid and nitric acids and forms boric acid.

$$2B + 3H_2SO_4 \longrightarrow 2H_3BO_3 + 3SO_2$$

$$B + 3HNO_3 \longrightarrow H_3BO_3 + 3NO_2$$

Boron reacts with fused sodium hydroxide and forms sodium borate.

$$2B + 6NaOH \longrightarrow 2Na_3BO_3 + 3H_2$$

Uses of boron:

1. Boron has the capacity to absorb neutrons. Hence, its isotope $^{10}B_5$ is used as moderator in nuclear reactors.
2. Amorphous boron is used as a rocket fuel igniter.
3. Boron is essential for the cell walls of plants.
4. Compounds of boron have many applications. For example eye drops, antiseptics, washing powders etc.. contains boric acid and borax. In the manufacture of Pyrex glass , boric oxide is used.

2.2.4. Borax [$Na_2B_4O_7.10H_2O$]:

Preparation:

Borax is a sodium salt of tetraboric acid. It is obtained from colemanite ore by boiling its solution with sodium carbonate.

$$2Ca_2B_6O_{11} + 3Na_2CO_3 + H_2O \xrightarrow{\Delta} 3Na_2B_4O_7 + 3CaCO_3 + Ca(OH)_2$$

XII_U2-P-Block.indd 33 04-01-2021 12:48:01